Variance Constraints Do Not Improve Prediction of Subsequent Insect Abundance Change: A Prospective Validation Across Long-Term Monitoring Studies

Elias De Jesús
Independent Researcher
ORCID: 0009-0007-0190-9143

Abstract

Higher-order variance structure has been proposed as a diagnostic of ecological vulnerability distinct from variance magnitude, but this proposition has not previously been tested prospectively. We report a prospective, pre-registered validation of that hypothesis using the van Klink et al. (2020) global database of long-term insect assemblage changes (70,955 abundance/biomass observations, 1,663 plots, 166 studies, 1925–2018). Working from 1,613 candidate, 1,129 eligible abundance series (99 studies; ≥10 usable years, chronologically split into an early predictor window and a held-out future window before any predictor–outcome association was examined), we tested three scale-invariant, higher-order constraint metrics — $D_{3/1}$, $D_{4/1}$, and $Q_{90/50}$ — as predictors of each series' subsequent abundance trend, using conventional baseline features and grouped, study-aware cross-validation. None of the three metrics showed the predicted direction after Holm correction (raw $p = 0.905$, 0.871, 0.043; Holm $p = 1.000$, 1.000, 0.171). Adding all three to a conventional baseline model did not improve out-of-sample prediction (pooled RMSE: $M_0 = 0.16900$, $M_1 = 0.16932$; $\Delta\text{RMSE} = -0.00032$; only 4 of 10 folds favored the augmented model; permutation controls $p = 0.454$ and $p = 0.583$). A separate, independently frozen prospective test of a study-level, hierarchical meta-variability statistic — $C_g = \mathrm{SD}_i(\mathrm{CV}_i)/\mathrm{Mean}_i(\mathrm{CV}_i)$, computed from each study's own first 10 historical years and evaluated against a fully held-out future period ($n = 11$ effective study contexts after a mandatory outcome-timing filter) — likewise showed no incremental predictive value (pooled RMSE: $M_0 = 0.141950$, $M_1 = 0.144303$; $\Delta\text{RMSE} = -0.002354$; 10,000-permutation one-sided $p = 0.5304$). Taylor's Law was independently, strongly replicated as a descriptive benchmark ($b = 1.930$, $R^2 = 0.974$), confirming the dataset itself is not featureless even though the tested constraint metrics carried no incremental prospective information. Constraint-like variance structure was measurable but did not provide incremental out-of-sample information about subsequent abundance change beyond conventional baseline features. These results support a conservative conclusion: descriptive constraint structure should not be interpreted as ecological vulnerability or early-warning information without prospective, target-specific validation.

Keywords: insect decline, variance structure, prospective validation, negative results, meta-variability, Taylor's Law, early-warning indicators

1 Introduction

Global insect populations have experienced substantial declines over recent decades, with estimates suggesting biomass reductions of 1–2% annually in some regions [Hallmann et al., 2017, van Klink et al., 2020a]. These declines carry profound implications for ecosystem function, including pollination services, nutrient cycling, and food web stability [Potts et al., 2010, Yang and Gratton, 2014]. Accurate assessment of population health and early detection of vulnerability are therefore important for conservation planning and ecosystem management.

Conventional monitoring approaches typically focus on two primary metrics: mean abundance trends and population variance. Mean trends capture directional change but may detect stress only after substantial population loss has occurred. Single-scale variance metrics, such as the coefficient of variation (CV), describe population stability but do not necessarily distinguish natural fluctuation regimes from pathological instability [Inchausti and Halley, 2002, Houlahan et al., 2007].

A growing body of literature has suggested that higher-order variance structure — the distribution and coupling of variabilities across populations, strata, and scales — may contain diagnostic information not captured by first-order statistics [Carpenter and Brock, 2006, Scheffer et al., 2009]. In particular, the variance of variances, or CV of CVs, has been proposed as a candidate early-warning indicator for critical transitions in complex systems [Dakos et al., 2012]. A companion idea is that ecological vulnerability may not concentrate where variance is largest: highly variable populations may possess intrinsic buffering capacity, while populations with more constrained variance may represent bottlenecks where perturbations amplify and propagate [Ives et al., 1999, Tilman, 1999].

Both ideas motivated an earlier, exploratory analysis of this same database, which reported a bounded population-level CV-of-CVs estimate, stratum-specific variance-structure contrasts, and proposed a family of candidate early-warning metrics. That analysis was explicitly diagnostic and correlational, and it called for validation on independent data and prospective time series before any of its candidate indicators could be treated as established. The present paper reports exactly that validation, conducted as a separate, prospectively frozen research program *after* the exploratory analysis, using a discipline in which every eligibility rule, predictor definition, outcome definition, validation design, negative control, and support criterion was committed to a version-controlled record *before* any predictor–outcome association was examined.

The central distinction this paper turns on is simple to state and easy to blur in practice:

> *A statistical feature may describe constrained variability without carrying information about a subsequent ecological outcome.*

Measurability is not predictive value. A metric can be well-defined, scale-invariant, and reproducibly estimable from real data — as all of the metrics tested here are — and still carry zero incremental information about what happens next in a specific series. Distinguishing these two properties requires holding out the future: computing every candidate predictor from an early historical window, and testing it only against a later, disjoint outcome window that the predictor never saw. This paper treats $D_{3/1}$, $D_{4/1}$, $Q_{90/50}$, and the contextual meta-variability statistic C_g throughout as *prospectively tested candidate indicators*, not as established vulnerability measures, and asks two falsifiable questions:

(1) Does a higher-order, scale-invariant constraint metric, measured in an early historical window, predict a series' subsequent abundance trend, beyond conventional baseline information?

(2) Does a study-level, hierarchical historical meta-variability statistic contain prospective information about subsequent abundance change, beyond the same conventional baseline?

2 Data and Methods

2.1 Data provenance and scale

The primary dataset is van Klink et al. [2020b] (Knowledge Network for Biocomplexity, DOI: 10.5063/F11V5C9V, CC BY 4.0), independently downloaded and checksum-verified against the provider's own reported checksums. The corpus contains **70,955** abundance/biomass observations across **1,663** plots from **166** original studies, spanning **1925–2018**, stratified into six ecological strata (Air, Water, Herb layer, Soil surface, Trees, Underground). These headline scale figures were independently re-verified against the raw, checksum-verified files rather than assumed from the source's own documentation.

The database aggregates heterogeneous monitoring designs: different studies used different trapping methods, sampling intensities, and reporting units. Raw abundance magnitudes are therefore **not** treated as directly comparable across studies or locations. The entire analysis instead focuses on within-series temporal structure (a series' own coefficient of variation, its own higher-order dispersion, its own temporal trend) and each series' own subsequent temporal change — quantities that are, by construction, invariant to a fixed multiplicative difference in reporting scale between studies (§2.4.1). Biomass records exist in the source (MetricAB =biomass) but are excluded from the primary analysis entirely; abundance (MetricAB =abundance) is the sole primary quantity, consistent with the frozen protocol's own scope.

2.2 Prospective design principle

Every choice reported in this section — eligibility, chronological split, predictor and outcome definitions, model specification, validation design, negative controls, multiplicity correction, and support criteria — was frozen and committed to a version-controlled protocol *before* any predictor–outcome association was computed. The research chronology is:

Commit	Content
`21e6992`	Round 0: protocol freeze (eligibility, outcome, metrics, model, controls)
`206742f`	Round 1: primary H1/H2/H3 results (executed once)
`d3246d9`	Round 2A: CV-of-CVs *feasibility* audit (no outcome touched)
`3e30670`	H4 protocol freeze (converts the feasibility recommendation into an immutable design)
`8d6db88`	H4: primary result (executed once)

2.3 Primary abundance-series construction and eligibility

The unit of analysis is a longitudinal series keyed by (DataSource_ID, Plot_ID, Stratum), restricted to abundance records. Of **1,613** candidate abundance series, **1,129** were *eligible*: ≥ 10 usable years (years with a non-null aggregated abundance value, summing multiple within-year sampling periods before any further computation), spanning **99** distinct studies.

Each eligible series' usable years were split chronologically into an early predictor window (the first $\lceil n/2 \rceil$ years) and a held-out future window (the remaining $\lfloor n/2 \rfloor$ years), guaranteeing ≥ 5 years in each window. No later-window observation was used in predictor construction at any point; this was enforced both structurally (the split is defined by ordering years, not calendar arithmetic) and by an explicit programmatic leakage check.

2.4 Predictors

Conventional baseline (M0). Five features computed from the early window alone: mean $\log(1 + \text{abundance})$, coefficient of variation, OLS temporal trend of $\log(1 + \text{abundance})$ on year, number of usable years, and temporal span.

Constraint metrics. For a series' early-window abundance values $\{x_1, \ldots, x_n\}$, define the power mean of order p,

$$M_p = \left(\frac{1}{n} \sum_{i=1}^{n} x_i^p \right)^{1/p}.$$

The three frozen constraint candidates are

$$D_{3/1} = \frac{M_3}{M_1}, \qquad D_{4/1} = \frac{M_4}{M_1}, \qquad Q_{90/50} = \frac{Q_{0.90}}{Q_{0.50}},$$

where Q_q denotes the q-quantile. By the power-mean inequality, $M_p \geq M_1$ for $p > 1$ with equality iff x is constant, so all three metrics attain their minimum value of 1 exactly at zero within-series dispersion (the "tightest" possible constraint) and increase with dispersion.

2.4.1 Scale invariance

For any $c > 0$, $M_p(c\,x) = c\,M_p(x)$, so $D_{3/1}$ and $D_{4/1}$ are invariant to multiplicative rescaling of a series' raw abundance; quantiles scale identically under $c > 0$, so $Q_{90/50}$ is invariant by the same argument. This property is what licenses comparing these metrics across studies recorded on different, unknown raw scales.

2.5 Validation design

Because series within a study share methodology, observers, and regional drivers, random row-wise cross-validation is invalid. All Round 1 inference used **10-fold GroupKFold**, grouped by DataSource_ID, so no study's series ever appear in both the training and test partition of a fold. Model M_0 uses the five baseline features; M_1 uses M_0 plus one (H1) or all three (H2) constraint metrics. Standardization is fit on the training fold only; a Ridge fallback ($\alpha = 1$) is used only if a training design matrix's condition number exceeds 10^8.

2.6 H1 / H2 and multiplicity

H1 (per metric): does the metric's early-window value predict the frozen direction of the later-window trend, individually, after study clustering? Evaluated via a cluster-robust (DataSource_ID) OLS coefficient. **H2**: does adding all three metrics to M_0 improve pooled out-of-fold RMSE ($\Delta\text{RMSE} = \text{RMSE}(M_0) - \text{RMSE}(M_1)$, positive favors the constraint metrics)? Four tests (H1-$D_{3/1}$, H1-$D_{4/1}$, H1-$Q_{90/50}$, H2) form one predeclared Holm family at $\alpha = 0.05$.

2.7 Negative controls

Three controls, 2,000 permutations each: **NC1** permutes each constraint metric within study; **NC2** permutes the future outcome within study; **NC3** reverses the early/late window roles as a leakage diagnostic (not a hypothesis test).

2.8 Prospective test of contextual meta-variability (H4)

A population-level CV-of-CVs is a single number and cannot, by itself, predict which individual series will decline more than another. A *prospective*, context-varying version requires a group-level, time-indexed statistic, computed from multiple contemporaneous series sharing a context. A separate feasibility audit (Round 2A, `d3246d9`) established, *without ever touching any outcome value*: LEVEL (a single historical estimate per context) was marginally feasible; CHANGE (a temporal slope of the statistic) was underpowered at every tested multiplicity (3–14 of 99 studies, depending on threshold); constituent-group sizes below 15 series showed materially unstable estimator behavior (95% bootstrap CI width ≥ 0.49 even at 30 constituent series, with a single extreme constituent CV still shifting the group estimate by ~ 0.33 on average); and the correct inferential scale is the (study, historical-window) pair, never the plot or series count broadcast onto it.

This feasibility recommendation was converted into an immutable protocol (`3e30670`) before any H4 result was computed. For a study's constituent series i with early-window abundance x_i,

$$\mathrm{CV}_i = \frac{\mathrm{SD}_i}{\mathrm{Mean}_i}, \qquad C_g = \frac{\mathrm{SD}_i(\mathrm{CV}_i)}{\mathrm{Mean}_i(\mathrm{CV}_i)}.$$

C_g is **contextual**: it is defined per study and is never treated as, or tested as, a universal constant. The historical window is the first 10 elements of the sorted, deduplicated *union* of usable years across a study's eligible series (rank-based, not calendar arithmetic); adequacy requires ≥ 15 constituent series with ≥ 2 observations in that window. Chronology follows Design C: this fixed 10-year window, then *all* remaining years as a fully held-out future period. A series may contribute an outcome only if its own Round 1 late window (unchanged formula, `ols_trend_slope`(log(1 + Number), Year)) falls strictly after the study's H4 cutoff — the unique condition required to keep the historical-to-future arrow honest, given that a series' own Round 1 split need not align with the study-level H4 cutoff.

Validation is **leave-one-study-out** (LOSO; the parameter-free generalization of GroupKFold once the number of eligible groups is much smaller than Round 1's 99). The primary statistic is pooled LOSO ΔRMSE = RMSE(M_0) − RMSE(M_1), with $M_1 = M_0 + C_g$. Inference uses a **between-study permutation** of the study-to-C_g mapping (the valid adaptation of Round 1's NC1/NC2 to a study-level, not series-level, predictor): 10,000 permutations, seed 20260907, one-sided, $p = (1 + \#\{b : \Delta_{\mathrm{perm},b} \geq \Delta_{\mathrm{obs}}\})/10001$. Direction was predeclared *before* any outcome was inspected, by direct analogy to Round 1's own H1 convention: a lower (tighter) historical C_g was predicted to predict a worse future trend (positive coefficient sign).

3 Results

3.1 Validation population

3.2 H1: individual constraint metrics

None of the three metrics showed the predicted direction after correction (Table 2). $Q_{90/50}$'s raw $p = 0.043$ must **not** be read as a near-significant positive result: its estimated direction was opposite the frozen prediction, and Holm correction leaves $p = 0.171$.

3.3 H2: incremental out-of-sample prediction

Adding $D_{3/1}$, $D_{4/1}$, and $Q_{90/50}$ to the conventional baseline did not improve, and slightly worsened, out-of-fold prediction (Table 3, Figure 1). Only 4 of 10 folds favored M_1. Both negative controls

Table 1: Dataset and validation population.

Quantity	Value
Total observations	70,955
Plots	1,663
Studies (source-level)	166
Year coverage	1925–2018
Candidate abundance series	1,613
Round 1 eligible series ($\geq$10 usable years)	1,129
Eligible studies	99
Common analysis sample (H1/H2)	1,126 series / 99 studies
H4 structurally adequate study-windows ($\geq$15 constituents)	16
H4 effective study N (after outcome-timing filter)	11

Table 2: H1 results: individual constraint metrics.

Metric	Predicted direction met?	Raw p	Holm p	Verdict
$D_{3/1}$	No	0.905	1.000	Not supported
$D_{4/1}$	No	0.871	1.000	Not supported
$Q_{90/50}$	No	0.043	0.171	Not supported

placed the observed ΔRMSE well inside their permutation null distributions (NC1 $p = 0.454$; NC2 $p = 0.583$; 2,000 permutations each). NC3 (time-reversal diagnostic) showed no leakage artifact. Holm-adjusted $p = 1.000$.

Table 3: Round 1 model performance, M_0 vs. M_1.

Model	Pooled out-of-fold RMSE	
M_0 (baseline)	0.16900	
M_1 (baseline + $D_{3/1}, D_{4/1}, Q_{90/50}$)	0.16932	
ΔRMSE = RMSE(M_0) − RMSE(M_1)	−0.00032	(folds favoring M_1: 4/10)

Adding $D_{3/1}$, $D_{4/1}$, and $Q_{90/50}$ did not improve prediction of subsequent abundance trends beyond the conventional baseline.

3.4 Secondary stratum analysis

Only the four strata passing the frozen adequacy gate ($\geq$30 eligible series and $\geq$5 independent studies) were inferentially tested: Air, Water, Herb layer, Soil surface. Trees and Underground failed the gate and were not tested. Direction was **not** consistent across strata, and constraint metrics made prediction of subsequent trends *materially worse* in two of the four:

- Herb layer: ΔRMSE ≈ -0.041
- Soil surface: ΔRMSE ≈ -0.051
- Air: ΔRMSE $\approx +0.0001$ (negligible)

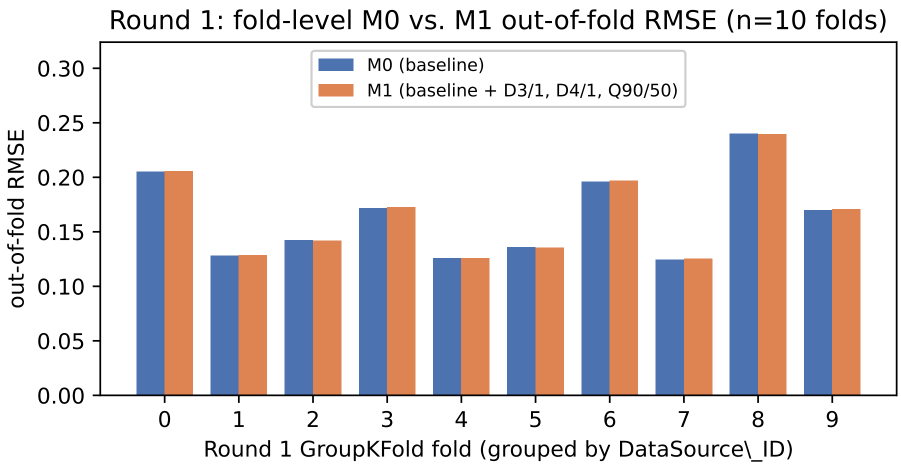

Figure 1: Round 1 fold-level out-of-fold RMSE, M_0 vs. M_1, across the 10 GroupKFold folds. Reproduced from `results/round1_primary_results.json` (commit `206742f`); no new analysis was run to produce this figure.

- Water: $\Delta\text{RMSE} \approx -0.0014$

This is a secondary, uncorrected result and is never promoted to the primary conclusion. It does **not** support treating Herb layer or Soil surface as validated vulnerability bottlenecks, nor Air or Water as validated buffers.

3.5 H4: prospective contextual meta-variability

Of 99 candidate studies, **16** were structurally adequate (a fixed 10-year historical window containing $\geq$ 15 constituent series). Applying the frozen outcome-timing filter — required so that every contributing series' future outcome genuinely postdates the study's historical cutoff — left an **effective study** $N = 11$. This is below the ∼19–20 studies estimated during the Round 2A feasibility audit; that estimate predated the outcome-timing filter, which was derived only when the H4 protocol itself was frozen, and $N = 11$, not $N \approx$ 19–20, is the number that supported the executed test. $N = 11$ clears the frozen floor of 10 below which the result would have been declared uninterpretable/blocked.

Across these 11 studies, C_g ranged from 0.2503 to 0.5606 (mean 0.4277, SD 0.0788, median 0.4056) — real, non-degenerate, contextual values, reported without removal, winsorization, or transformation.

Seven of 11 LOSO folds individually favored M_1, yet pooled performance still favored M_0 (Table 4): a majority of folds showing a small favorable shift does not override pooled error when one fold (a single study) carries a large unfavorable shift ($\Delta\text{RMSE}_{\text{fold}} \approx -0.063$ for that study alone). The secondary standardized C_g coefficient had the predicted sign but was not significant under asymptotic cluster-robust inference; per the frozen protocol, this secondary result cannot rescue a failed primary criterion. The 10,000-permutation test (Figure 2) placed the observed ΔRMSE near the null median (null mean −0.00317, SD 0.00474; $q_{05} = -0.01267$, $q_{50} = -0.00208$, $q_{95} = 0.00320$), giving $p = 0.5304$ — far from the frozen $\alpha = 0.05$ threshold in either direction.

Frozen verdict: NOT SUPPORTED.

Table 4: H4 prospective result.

Quantity	Value
Effective study N	11
RMSE(M_0)	0.141950
RMSE(M_1)	0.144303
ΔRMSE	-0.002354
LOSO folds favoring M_1	7 / 11
Secondary standardized C_g coefficient	+0.00875 (correct sign; cluster-robust $p = 0.4423$)
Permutation p (10,000 draws, one-sided)	0.5304

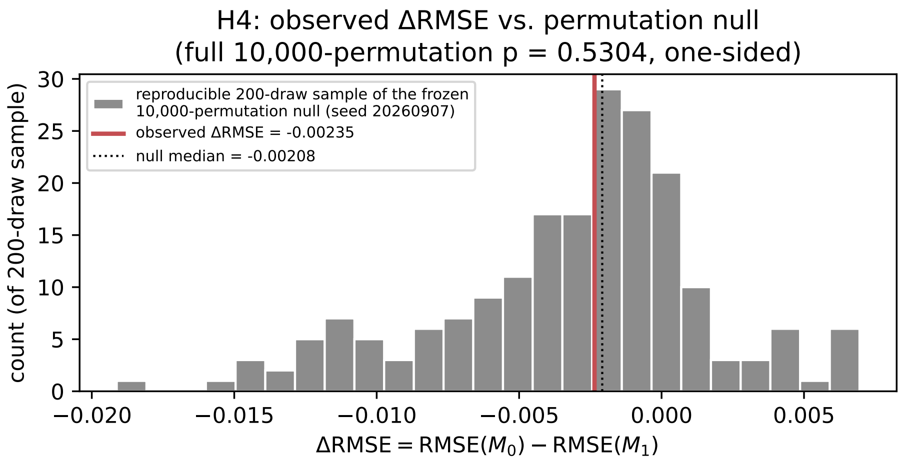

Figure 2: H4: observed ΔRMSE against a reproducible 200-draw sample of the frozen 10,000-permutation null distribution (full summary statistics computed from all 10,000 permutations; commit `8d6db88`). No new permutation analysis was run to produce this figure.

3.6 Descriptive benchmarks

3.6.1 CV-of-CVs

The exploratory manuscript this validation program tests reported a population-level $\mathrm{CV}_{\mathrm{CVs}} \approx 0.610$. An independent recomputation under this project's own transparent, disclosed methodology gave **0.512** — in the same general range, but not an exact reproduction (the original code and exact filtering choices were not available to this project). Neither value is treated as universal. A finite empirical CV-of-CVs establishes only finite dispersion in the specific population it was computed from; it does **not**, by itself, establish an ecological bound, a sampling artifact, or predictive value — these are four logically distinct claims, and only the first follows automatically from a finite number. The prospective H4 test above did not demonstrate that historical, contextual CV-of-CVs improves prediction of subsequent abundance change. **Measurability is not universality, and descriptive structure is not predictive information.**

3.6.2 Taylor's Law

Log–log regression of within-series variance against within-series mean across eligible series gave $b = 1.930$, $R^2 = 0.974$ — close to, though not assumed identical to, the exploratory manuscript's reported $b \approx 1.959$, $R^2 \approx 0.962$. This is a strong, independently reproduced descriptive regularity, confirming the dataset is not featureless. It is **not** evidence that the tested constraint metrics predict vulnerability, and no time-varying "Taylor exponent shift" statistic was ever defined or tested in this program; any such claim in the exploratory manuscript remains untested.

4 Discussion

The predeclared higher-order variance metrics — $D_{3/1}$, $D_{4/1}$, $Q_{90/50}$, and the independently frozen contextual statistic C_g — were all measurable, reproducible, and scale-invariant by construction. None added incremental prospective information about subsequent abundance change beyond conventional baseline features, under either of two independently designed, differently scaled prospective tests. This is the central result of this paper.

4.1 Measurable structure versus target-specific information

A recurring temptation in this literature is to read descriptive structure — a tight variance regime, a bounded CV-of-CVs, a particular Taylor exponent — directly as an ecological signal. The results here illustrate why that step needs its own evidence: *structural tightness alone is insufficient evidence of ecological vulnerability.* This is not a universal theorem about all possible constraint metrics in all possible systems; it is what was found for the three specific, predeclared metrics and the one specific, predeclared contextual statistic tested in this specific dataset and design.

4.2 Why the null results matter

A negative result under a design this disciplined is informative, not merely an absence of finding. It shows that a descriptive pattern observed in one analysis (the original, exploratory stratified variance contrasts) does not automatically survive being asked a sharper, prospective question of the same data. It highlights a hierarchical effective-sample-size problem (§4.3) that a purely descriptive analysis of the same corpus could not have exposed. It demonstrates, concretely, why chronological, study-aware validation is stricter than cross-sectional comparison: Round 1's own grouped cross-validation and H4's leave-one-study-out design are both engineered specifically to prevent the kind of pseudoreplication that a naive plot-level or observation-level analysis of this corpus would risk. And the negative controls (NC1–NC3, and H4's between-study permutation) directly support the interpretation that there is no detectable incremental signal in the tested metrics, rather than merely an absence of statistical significance under one particular model.

4.3 Effective sample size

The database contains 70,955 observations and 1,663 plots. H4's final inferential test ultimately rested on **11** effective study contexts. This is not a contradiction or a design flaw — it is the direct, unavoidable consequence of testing a *hierarchical, study-level* predictor: C_g does not vary within a study, so no amount of within-study replication (plots, series, observations) increases the number of independent points at which C_g itself varies. Large raw observational N does not imply large inferential N for a contextual predictor of this kind. This is a substantive methodological finding in its own right, distinct from the negative result itself: a future study proposing a contextual, hierarchical early-warning statistic should budget its expected effective N at the level the predictor actually varies at, not at the level the raw corpus happens to report. The H4 test reported here should accordingly be read as *prospectively specified but small-effective-N*, not as strongly powered; failure to support H4 is evidence against a detectable predictive effect *under this specific design*, not proof that no ecological relationship of this kind could ever exist at a larger, more richly sampled effective N.

5 Limitations

- The source database aggregates heterogeneous monitoring designs (sampling methods, units, durations) across originally independent studies.
- All analyses are observational; no causal claim is made or licensed by any result in this paper.
- Subsequent abundance trend is one operationalization of ecological vulnerability among several plausible ones; a null result for this outcome does not test other outcomes (e.g. local extinction, biomass decline, food-web disruption).
- Round 1's effective study N (99) and H4's effective study N (11) differ by an order of magnitude, for the structural reasons given in §4.3.
- The C_g estimator showed material single-value sensitivity even at its largest tested constituent-group sizes (Round 2A).
- Study structure is unequal: which studies meet H4's adequacy gate is itself correlated with a study's own plot count, a design artifact rather than an ecological one (Round 2A).
- No causal inference, no claim of a universal CV-of-CVs constant or ecological bound, no real-time forecasting validation, and no independent external-dataset validation are made anywhere in this paper.

These limitations qualify the scope of the negative result; they do not undermine its validity. The result stands as: under this specific, prospectively frozen design, on this specific dataset, these specific metrics did not add prospective predictive value.

6 Future Work (untested extensions)

The exploratory manuscript this program validates included two cross-domain illustrations that are **not** part of, and do not provide evidence for, the primary result above.

Bioacoustic classification data (Cicadidae/Orthoptera species classification samples) were never incorporated into any frozen protocol in this validation program and remain an **untested** extension — a plausible future automated-monitoring application, not validating evidence.

Paleontological (Paleozoic chondrichthyan) cross-scale comparison was likewise never tested in this program and remains an **untested** cross-scale analogy. Neither extension is treated as confirming, illustrating, or generalizing the primary result in any of the sections above.

7 Conservation implications

The prospective validation reported here does **not** support directing conservation attention toward Herb layer, Soil surface, or Underground strata on the basis of the tested constraint metrics, nor toward Air or Water strata as validated buffers (§3.3). A more defensible statement, given the evidence in this paper, is: variance-structure metrics of the kind tested here should not be used for conservation prioritization without independent, prospective, target-specific validation.

8 Epistemic Status

Supported: the source dataset's scale and provenance; the eligible validation populations (Round 1: 1,129 series/99 studies; H4: 11 effective study contexts); strong Taylor's Law scaling as a descriptive benchmark; reproducibility of every frozen analysis (123 tests passing at the final H4 state); contextual measurability of CV-of-CVs (as C_g, a real, computable, per-study quantity).

Not supported: $D_{3/1}$, $D_{4/1}$, and $Q_{90/50}$ as vulnerability predictors; incremental predictive advantage of the Round 1 constraint metrics over conventional baseline features; contextual C_g's prospective early-warning value; the Herb-layer/Soil-surface bottleneck interpretation; the Air/Water buffer interpretation; a universal CV-of-CVs constant or ecological bound.

Underpowered: temporal CHANGE in CV-of-CVs (3–14 of 99 studies depending on threshold; excluded from H4 by design).

Untested: any causal mechanism; the bioacoustic extension; the paleontological extension; real-time forecasting utility; generalization to any external, independently collected dataset.

9 Conclusion

$D_{3/1}$, $D_{4/1}$, and $Q_{90/50}$ did not improve prospective prediction of subsequent insect abundance change beyond a conventional baseline. An independently, separately frozen prospective test of contextual historical CV-of-CVs likewise failed to demonstrate incremental predictive value. Taylor's Law scaling remained a strong, independently reproduced descriptive feature of the same dataset, confirming that the null results above are not an artifact of an otherwise unstructured corpus. Descriptive constraint structure should not be interpreted as ecological vulnerability without demonstrated, target-specific predictive information.

These results support a conservative distinction between descriptive variance structure and ecological predictive information. Constraint-like statistical organization may be measurable without functioning as an early-warning indicator, and its ecological significance should therefore be established prospectively rather than inferred from structural tightness alone.

10 Reproducibility and Data Availability

All frozen protocols, execution code, tests, and results referenced in this paper are version-controlled in the public GitHub repository, *insect-variance-constraint-validation*[1], the living, version-controlled computational record for this work. A permanent, citable archival deposit of this manuscript is separately held on Zenodo[2] (De Jesús, 2026); the Zenodo DOI is the version of record for citation purposes, while the GitHub repository is the record to consult for the current state of the

[1] `https://github.com/innerlightr-wq/insect-variance-constraint-validation`

[2] `https://doi.org/10.5281/zenodo.22646397`

code, tests, and results. The commit that produced this manuscript revision is `f37a7b3`, itself built entirely from the frozen, already-executed research chronology below (`f37a7b3` is a post-hoc manuscript-writing commit, not part of that pre-result freeze sequence). The full research chronology:

`21e6992`	Round 0 protocol freeze
`206742f`	Round 1 results
`d3246d9`	CV-of-CVs feasibility audit
`3e30670`	H4 protocol freeze
`8d6db88`	H4 results

123 tests were passing at the final H4 state referenced above. This count is reported as reproducibility information only, not as scientific evidence for or against any hypothesis in this paper.

The primary dataset [van Klink et al., 2020b] is licensed CC BY 4.0; it is not redistributed by this repository (raw third-party data is never committed, per repository policy), but is fully re-downloadable and checksum-verifiable from its original source using the commands and checksums recorded in the repository's own data-provenance documentation. This repository uses a dual license for its own original content: original software/code is released under the MIT License, and original manuscript text, documentation, and figures (including this manuscript) are released under the Creative Commons Attribution 4.0 International License (CC BY 4.0). Neither license extends to, or relicenses, the third-party insect monitoring dataset above, which remains governed solely by its own original CC BY 4.0 terms as an independent, externally authored dataset.

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